The Gender Pay Gap and its Determinants across the Human Capital Distribution

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Abstract

This paper leverages a unique linkage between American Community Survey data and postsecondary transcript records to examine how the gender pay gap, and its proximate determinants, varies across the distribution of education credentials in the 15 years following graduation. Although recent literature focuses on career disparities between the highest-educated women and men, we find evidence that the pay gap is smaller at higher education levels. Field-of-degree and occupation effects explain most of the gap among top bachelor's graduates, while labor supply and unobserved channels matter more for less-competitive bachelor's, associate's, and certificate graduates. This heterogeneity in gap levels and mechanisms is especially large in the first decade following graduation. Our results suggest that contemporary early-career gender inequality lacks a unified explanation and requires different policy interventions for different subgroups. More research is needed to understand the larger unexplained gender pay gap among less-educated individuals.

Keyword*:* gender pay gap, postsecondary transcript records, college selectivity, human capital, field and occupation choice, labor supply

JEL Classification: I24, I26, J16, J31

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I. Introduction

In 2022, the average woman in the U.S. workforce earned 18 percent lower hourly wages than the average man (Pew Research Center, 2023). A consistent finding from the literature is that the average gender pay gap cannot be explained by gender differences in human-capital variables such as educational attainment (see the reviews of Altonji and Blank, 1999; Bertrand, 2011; and Blau and Kahn, 2017). Recent literature has studied gender inequality in subpopulations with uniquely high levels of education: e.g. graduates from top universities, or employees in top-paying occupations that require advanced degrees (Noonan et al. 2005; Bertrand et al. 2010; Matsa and Miller 2011; Goldin 2014; Bursztyn et al. 2017; Fernandez and Campero 2017; Wiswall and Zafar 2018; Cortés et al. 2021; Foster et al. 2022; Wasserman 2023). This literature has generated important findings, but it is unclear whether these findings apply to other segments of the population. We take up that question in this paper by analyzing how the contemporary gender pay gap varies across the distribution of education credentials.

Rising U.S. inequality and diversification of the higher education system (Gerber and Cheung 2008; Hoxby 2009; Lovenheim and Smith 2022) suggests a need to move beyond the average gender pay gap to an analysis of how the gap varies across different human-capital and labor-market contexts. On the one hand, high returns to specialized professional skills (Deming and Kahn 2018; Deming and Noray 2020), high returns to intensive hours in certain professions (Goldin 2014), and the male-dominated networks that often persist in these environments, may widen gender inequality at the top of the human capital distribution. On the other hand, exposure to unpredictable shift-scheduling (Schneider and Harknett 2019), lack of employer-provided work-family support, and tenuous childcare arrangements and social-support networks (Su and Dunifon 2017; Harknett et al. 2020; Cortés and Pan 2020) may widen gender inequality at the lower end of the distribution (Binder 2021). While some research has investigated the relationship between overall earnings inequality and the average gender pay gap (e.g. Blau and Kahn 1997, 2000; Mulligan and Rubinstein 2008; Fortin et al. 2017), little work to our knowledge examines gender inequality across an unequal distribution of skills and opportunities.

We develop a dataset that is uniquely qualified for such an analysis by linking 2005-2019 American Community Survey respondents to their postsecondary transcript records, drawing on a partnership between the U.S. Census Bureau and state higher education systems. We observe information on field of degree, occupation, labor supply, earnings, and family structure for a large sample of 2003-2013 graduates, whose credentials range from short-term certificates to bachelor's degrees conferred by highly-selective American colleges. Ours is the only U.S. dataset of which we are aware with the information and statistical power necessary to estimate the gender pay gap, and its proximate determinants, across such a range of credentials.[1] The dataset allows us to examine labor market outcomes up to 15 years following graduation. Research has shown that the gender pay gap increases substantially during this time and is relatively stable thereafter (Manning and Swaffield 2008). These early-career dynamics are the subject of their own literature (discussed in Section VI), but they also inform gender inequality over individuals' full careers.

[1] Other work, albeit without an explicit focus on gender inequality, has leveraged administrative data to study the returns to different types of postsecondary credentials in Norway (Kirkeboen et al. 2016; Bertrand et al. 2021).

Our analysis generates several findings regarding contemporary gender inequality. First, by one measure, the gap is constant across the distribution of education credentials; by another, the gap falls as degree level rises.[2] Second, field-of-degree sorting explains a substantial portion of the gap at higher levels, but is a less-important determinant of the gap at lower levels. Third, field-conditional occupation sorting is an important determinant of the gap at all levels. Fourth, occupation-conditional labor supply differences are an important determinant of the gap at lower degree levels, but a minor determinant at the top. Fifth, in the years following graduation, the gap is smallest at the highest levels, but after 15 years of labor-market exposure, the pay gap is similar across degree levels. Sixth, the effects of children on the gender pay gap are similar at both certificate and top bachelor's levels, although they operate primarily through labor supply and residual channels for the former, and through occupation and labor supply channels for the latter. Finally, we confirm the negative relationship between gap magnitude and degree level in a subsample of Business graduates, and find uniquely high contributions of occupation effects to the gap among top Business graduates.

Our results indicate that gender pay inequality is a pervasive phenomenon throughout the labor market for recent graduates, but also that different mechanisms are important at different positions within the market. We discuss how some of our headline estimates relate to prior estimates in the early-career gender gap literature. This discussion suggests lower gender inequality in the modern cohorts we consider (relative to the less-recent cohorts studied by the literature), as well as a lower (but still first-order important) level of field and occupation segregation among bachelor's graduates. It thus appears that gender inequality continues to fall, albeit slowly. At the same time, remaining inequality eludes a unified explanation across the different contexts faced by differently-trained postsecondary graduates.

The paper proceeds as follows. In the next section, we describe the construction of our dataset and some basic summary statistics. Section III describes our decomposition methodology. Section IV presents and interprets our main gender pay gap decompositions. Section V repeats selected analyses from Section IV on a subsample of Business graduates. Section VI compares our findings to those of related prior studies. Section VII offers concluding remarks and policy implications.

II. Data

We sourced postsecondary transcript records from a subset of the state higher education systems that participate in the Post-Secondary Employment Outcomes (PSEO) partnership:[3] specifically, Connecticut State Colleges and Universities, the City University of New York, the Louisiana Board of Regents, the Missouri Department of Higher Education and Workforce Development, the Ohio Department of Higher Education, the State Council of Higher Education

[2] Our paper estimates *conditional average* gender gaps across various levels of pre-labor-market human capital (i.e. across various points of the *potential earnings* distribution). Others have estimated *unconditional percentile* gender gaps across various points of the *observed earnings* distribution (e.g. Arulampalam et al. 2007, Blau and Kahn 2017). Our results show a contrast between the two approaches: while we find a smaller gap at higher levels of observable human capital, these studies report a larger gap at higher percentiles of observed earnings. While further work is needed to explore these contrasting patterns, one proximate (certainly incomplete) explanation is that the preponderance of top male earners is not well-predicted by observable human capital and occupation choices.

[3] For more information about the PSEO data, go to: https://lehd.ces.census.gov/data/pseo_experimental.html

for Virginia, the State University of New York, the Texas Higher Education Coordinating Board, and the Utah System of Higher Education. These systems cover all four regions of the United States. The transcripts we observe are person-degree[4] records that include information on institution, degree type (short-term certificate, 1-2-year certificate, associate, or bachelor's), degree field, and date of graduation. Each graduate is assigned a Protected Identification Key (PIK) by the U.S. Census Bureau's Person Identification Validation System (PVS).[5] These PIKs allow the transcript data to be matched with person-level responses to the American Community Survey (ACS). The ACS provides key information on earnings, occupation, and labor supply, as well as demographic information on family structure. We observe annual transcript data from most systems beginning in the early 2000s, and ACS data for survey years 2005-2019.

A. Main Sample

We used this information to construct a dataset based on the following criteria: individuals who graduated in 2003-2013, who were between the ages of 19 and 42 at graduation, who were between the ages of 19 and 59 when they were sampled by the ACS, and who graduated prior to being sampled by the ACS. Thus, each graduating cohort can be observed for a minimum of six and maximum of 16 years following graduation. Table 1 records sample counts and cohort coverage by higher education system. Our sample contains approximately 630,000 graduates, nearly 30% of which graduated from an institution in Texas. Approximately 108,000, or 17% of the sample, graduated from an Ohio institution, another 107,000 graduated from a SUNY institution, and another 93,000 (15%) graduated from an institution in Virginia. The other five systems comprise the remaining 21% of the sample.

TABLE 1. STATE HIGHER EDUCATION SYSTEMS IN OUR SAMPLE

System	PSEO	Our Sample	
	Coverage	Coverage	N
Connecticut State Colleges and Universities (CSCU)	2001-2020	2003-2013	14,500
City University of New York (CUNY)	2001-2019	2003-2013	38,500
Lousiana Board of Regents (LABOR)	2001-2019	2003-2013	35,500
Missouri Department of Higher Education and Workforce Development (MDHEWD)	2005-2020	2005-2013	37,500
Ohio Department of Higher Education (ODHE)	2002-2020	2003-2013	108,000
State Council of Higher Education for Virginia (SCHEV)	1997-2020	2003-2013	93,000
State University of New York (SUNY)	2002-2019	2003-2013	107,000
Texas Higher Education Coordinating Board (THECB)	2000-2019	2003-2013	185,000
Utah System of Higher Education (USHE)	2010-2020	2010-2013	11,500
Total			630,000

Source: PSEO transcript records linked to the 2005-2019 ACS.
Note: Ns rounded for disclosure avoidance purposes.

[4] If an individual earns multiple degrees, there is an observation in the data for each degree earned. If an individual earns one degree, but has two or more majors, this individual will have an observation for each major. Likewise, graduation from multiple institutions will lead to multiple observations. We have verified that negligible shares of our sample appear in the PSEO data as both a sub-baccalaureate degree earner and a baccalaureate degree earner.
[5] For more information on PVS, see Wagner and Layne (2014).

Table 2 presents selected summary statistics of our sample. Women comprise nearly 59% of observed graduates, reflecting known gender differences in degree attainment in the United States and other advanced countries (Goldin et al. 2006; Van Bavel et al. 2018). Most individuals in our sample are in their 30s. The average individual graduated in 2008 and is observed six years later, during the 2014 ACS data year. Most individuals are White and non-Hispanic, although we observe substantial numbers of Black non-Hispanic, Asian non-Hispanic, and Hispanic individuals as well. The majority of graduates earned a bachelor's degree, while 31 percent earned an associate degree and the remaining 9 percent obtained a certificate.

Table 2 also records summary information on earnings over the 12 months prior to each individual's ACS interview. The average woman in our sample earned $38,700, while the average man earned $53,200, leading to a gender gap of $14,500, or 27.3 percent of the average man's earnings. We find a similar gap of $13,600, or 26.4 percent of the average man's earnings, when we consider wage and salary earnings only. Despite this substantial gap, there are small differences in labor force attachment: 83 percent of women, as compared with 88 percent of men, earned at least $5,000.

Due to the select geographic and institutional coverage of our sample, it is hard to name a target population that our sample can be considered to represent. However, we achieved reasonable comparisons between our sample and a nationally representative sample of similarly-aged individuals with college education drawn from the 2016 ACS 5-year public-use file. Our sample contains a slightly larger share of individuals identifying as White non-Hispanic and exhibits a slightly smaller baseline average gender pay gap. Appendix Table A1 provides further information.

TABLE 2. SUMMARY STATISTICS OF MAIN SAMPLE

Panel A. Demographics			Panel B. Human Capital and Earnings		
	Mean	SD		Mean	SD
Woman	0.59	0.49	Certificate	0.09	0.28
Age at graduation	25.2	5.0	Associate	0.31	0.46
Age	31.3	6.4	Bachelor	0.61	0.49
Graduation year	2008	3.1	Selective Bachelor	0.23	0.42
ACS year	2014	3.7	Total earnings, women	38.7	39.3
White, non-Hispanic	0.71	0.45	Total earnings, men	53.2	56.2
Black, non-Hispanic	0.09	0.28	Wage/sal earnings, women	38.0	38.6
Asian, non-Hispanic	0.05	0.22	Wage/sal earnings, men	51.5	53.9
Other, non-Hispanic	0.03	0.17	Earned < $5000, women	0.17	0.37
Hispanic	0.12	0.32	Earned < $5000, men	0.12	0.32
N = 630,000					

Source: PSEO transcript records linked to the 2005-2019 ACS and the 2004 Barron's Guide.
Note: Sample membership described in Section II.A. "SD" stands for standard deviation. "Selective Bachelor" corresponds to students who graduated from an institution rated as Very Competitive, Highly Competitive, or Most Competitive according to Barron's 2004 guide. Earnings are for the 12 months prior to when the individual responded to the ACS. Earnings are inflated to 2019 prices using the CPI for all urban consumers and are expressed in thousands.

To distinguish among the wide variety of bachelor's programs in the United States and the investments they make in their students, we merged 2004 Barron's Competitiveness ratings for bachelor's programs onto our dataset at the institution level. Barron's rates programs with competitive admissions policies (i.e. that are not open enrollment), and applies a 1-7 rating scale to the programs it rates. We recoded this to a 1-6 scale, pooling the two least competitive categories (which were small), and assigned a seventh category to programs left unrated by Barron's. This category is small, accounting for under three percent of the full sample.

Table 3 presents student-weighted average institutional characteristics by Barron's category, along with student and institution counts. We sourced institutional characteristics from the 2004-05 edition of the U.S. Department of Education's College Scorecard database.[6] The table shows substantial stratification in institutional selectivity, investments, and graduation outcomes across the Barron's categories. For example, while nearly two-thirds of applicants to Most Competitive institutions in our sample are rejected, only 4.6 percent of applicants to Non-Competitive institutions are rejected. Most Competitive institutions spend over twice as much on instruction as do the Less and Non-Competitive institutions. While under 30 percent of students at Non-Competitive institutions graduate in four years, over 90 percent of those at Most Competitive institutions do. Moreover, male graduates from Most Competitive institutions earn 63.2 percent more income on average than do male graduates from Non-Competitive institutions.

TABLE 3. AVERAGE INSTITUTIONAL CHARACTERISTICS BY BARRON'S CATEGORY

Condensed Barron's Rating	Rejection Rate	Expenditures / FTE student	Graduation Rate (4-year)	Male Total Earnings	N Students	N Institutions
Non-Competitive	0.046	4,960	0.294	52,200	36,500	25
Less Competitive	0.188	4,660	0.335	52,100	36,000	22
Competitive	0.301	5,700	0.468	55,500	150,000	64
Very Competitive	0.331	7,180	0.606	62,900	75,500	22
Highly Competitive	0.418	7,570	0.717	71,400	54,000	9
Most Competitive	0.625	10,040	0.903	85,200	12,500	6

Source: PSEO transcript records linked to the 2005-2019 ACS, 2004 Barron's Guide, and 2004-05 College Scorecard.
Note: Expenditures are instructional expenditures per full-time-equivalent (FTE) student, expressed in Fiscal Year 2004 dollars. Male earnings are for the 12 months prior to when the graduate responded to the ACS and are expressed at 2019 prices. Expenditures are rounded to the nearest ten, male earnings are rounded to the nearest hundred, and sample counts are rounded according to internal rounding rules for disclosure avoidance purposes.

Thus, in total, we consider ten different postsecondary *degree levels* in our analysis: short-term certificate, one- or two-year certificate, associate, and the seven levels of bachelor's degrees described above.[7]

[6] The College Scorecard is produced by the Department of Education and contains information on all institutions that participate in federal aid programs. This tends to cover the near-universe of bachelor's degree programs (certainly those rated by Barron's), but with some gaps in coverage for specialized bachelor's programs. It combines data from the Integrated Postsecondary Education Data System, the National Student Loan Data System, and administrative tax records from the Department of the Treasury to provide data at the institution level and by field of study on outcomes like admissions, enrollment, costs, completion rates, average earnings, and loan repayment rates. The data are public and can be accessed here: College Scorecard | College Scorecard (ed.gov).

[7] To measure the extent to which graduates receive additional education following their observation in the transcript data, we provide Appendix Table A2, which restricts our main sample to individuals observed at least two years

B. Subsample to Investigate Effects of Children

To investigate how the presence of children affects the gender pay gap and its underlying mechanisms, we used PIK assignments to merge our main sample to the Census Household Composition Key (described in Genadek et al. 2021). The CHCK file links parents to children based on birth certificate information recorded on the Social Security Administration's Numident file and covers births from 1997-2021. The key advantage of this dataset over the ACS is that the latter can only identify parent-child linkages based on the relationship to household head variable. This makes it difficult to associate parents with children in certain multi-family or multi-generational households without unverifiable assumptions, and impossible to associate parents with children who are not coresidents. Moreover, the frequencies of such living arrangements potentially vary across the postsecondary human capital distribution. Using the CHCK eliminates this complication, but at the cost of only being able to observe births since 1997. This means that we cannot observe complete birth histories for individuals that had children at an unusually young age and/or earned a postsecondary degree at an unusually old age.

To avoid these censoring issues, we created a subsample of individuals who responded to the ACS during the 2016-2019 survey years and who were no older than 37 when they responded. We also imposed a minimum age limit of 28 to obtain a sample of individuals observed at ages when fertility rates are relatively high. This subsample is about one-fourth as large as the main sample, but we verified that key gender pay gap patterns were quite similar between the two samples. Using the PIK-based linkage to the CHCK file, we created a binary variable taking the value 1 if the individual had a child at least 6 months prior to the survey year (since the ACS elicits information on earnings and labor supply over the 12 months prior to the survey week).

III. Methodology

We consider two different earnings concepts throughout this paper: total real earnings (including zeros), and the natural logarithm of real earnings from wage and salary employment.[8] We view the two measures as complementary, and do not take a stance on which is preferred. For example, the first measure does not restrict on any definition of formal employment, and thus is as inclusive as possible of differences in paid labor supply. However, it may be sensitive to outlier

following receipt of their PSEO degree, and records educational attainment information reported to ACS by PSEO degree level. Around 20 percent of individuals who we observe earning a certificate degree in the PSEO report having attained a bachelor's degree in the ACS, and 44 percent of PSEO associate graduates report having a bachelor's degree. Moreover, almost half of Most Competitive bachelor's graduates report an advanced degree (compared to around 22 percent among graduates from Non-Competitive or Unrated bachelor's programs, and 8 percent among associate's graduates). One might argue that this phenomenon results in misclassification of our education categories: if so, then our approach is biased *against* finding heterogeneity in the gender gap across degree levels. Nevertheless, comparisons of male postgraduation earnings across PSEO degree levels suggest that our approach produces substantial stratification: the average male graduate with an associate (certificate) degree earned $43,700 ($39,700), or roughly one-half as much as the average male graduate from a Most Competitive bachelor institution. In general, we interpret the attainment of additional education as part of the natural school-work paths our graduates may take following initial postsecondary graduation; these paths, to the extent that they differ by gender, will be reflected in our estimates.

[8] Real earnings are expressed in US dollars at 2019 prices.

earners or to transitory swings in self-employment earnings. On the other hand, the second measure excludes self-employment earnings and those without any labor-force attachment, but this results in a comparison of women and men with similar levels of participation in the formal workforce. The log scaling also belongs to the class of Mincer earnings functions commonly used in the labor economics literature.

A. Explanatory and Stratifying Variables

We estimate a series of earnings models in terms of several predictor variables. We consider two different types of variables. Our main variables of interest are *explanatory* variables, i.e., proximate mechanisms of the gender pay gap. We allow for heterogeneity in our estimates by considering *stratifying* variables, wherein we estimate a separate model for each stratum.

- ***Explanatory*** variables:
    - *Field of degree fixed effects*. The PSEO transcripts record six-digit field codes from the Classification of Instructional Programs (CIP). Many of these codes contain small numbers of students or are redundant. To avoid small cells, we aggregated these codes to the two-digit level. Although two-digit codes are the broadest taxonomy of field of study, they still contain detail: we observe 44 distinct codes in our dataset.[9] Moreover, as a robustness check, we reran our main analyses with four-digit codes and found broadly similar results, as we discuss in the next section.
    - *Occupation and industry fixed effects*. We aggregated the 535 ACS occupation codes into 316 different labor market sectors, following the guidance of Foster et al. (2021).[10] We also aggregated ACS industry codes into 18 major industry codes, related to those commonly used in studies of gender segregation in the labor market (e.g. Schaller at al. 2019).
    - *Labor supply*. To allow for non-linear relationships between work intensity and earnings (Goldin 2014), we code labor supply as a flexible categorical variable rather than simply as the number of hours worked during the year. Specifically, we interact six different "weeks worked" bins (1-13, 14-26, 27-39, 40-47, 48-49, 50-52) with eight different "usual hours worked per week" bins (1-20, 21-30, 31-39, 40, 41-49, 50-54, 55-64, 65-99).

- ***Stratifying*** variables:
    - *Degree level*. We estimate separate models for graduates with each of the following degrees: short-term certificate, one- or two-year certificate, associate, bachelor's from an institution not rated by Barron's; bachelor's from each of the six Barron's categories of institutions described in Table 3. For ease of

[9] Example two-digit CIP codes include Education; Engineering; Biological and Biomedical Sciences; Mathematics and Statistics; Physical Sciences; and Psychology. For more details on CIP codes and a full list of the taxonomy, see: https://nces.ed.gov/ipeds/cipcode/

[10] We thank Brad Foster for generously sharing the occupation crosswalk used in Foster et al. (2021), which also appears as Appendix Table A2 of that paper.

exposition, we sometimes aggregate these ten levels to a smaller number or focus on a comparison of certificate degree holders to Most Competitive bachelor's degree holders.

- *Time since graduation.* To measure how the gender pay gap evolves after degree completion, some analyses group graduates into four different time bins (0-3 years since graduation, 4-7 years, 8-11 years, 12-16 years) and estimate earnings models separately for each bin.
- *Parental status*. Some analyses use the CHCK subsample to classify graduates as childless or as parents and estimate separate earnings models for each status. The difference in the results for parents versus non-parents yields a descriptive estimate of how parenthood affects the gender pay gap and its explanatory mechanisms.

B. Sequential Decomposition Framework

Within each stratum that we consider, we estimate the total gender pay gap; the portion attributable to each explanatory variable; and the residual portion. One approach would be to estimate a single earnings regression with all covariates included and compute the standard Oaxaca-Blinder decomposition or a related variant. This approach assumes that covariates are not endogenous: observed correlation between two covariate groups X_1 and X_2 is incidental and not because X_1 causally relates to X_2. As shown by Gelbach (2016), this "full model" partials out the correlation between X_1 and X_2, and the X_1-conditional wage effect of X_2, to estimate the wage effect of X_1. This estimated partial effect is then used to compute the contribution of cross-group differences in X_1 to the observed wage gap.

We propose an alternative approach that strikes us as more appropriate for our current setting. To illustrate, let X_1 refer to field of degree and X_2 to occupation. Degree field relates to occupation in a plausibly causal manner: if a graduate's field of degree were manipulated, one would expect a *ceteris paribus* change in their skills, preferences, and eventual occupation—and this shift in occupation *in itself* would induce a potential earnings response.[11] This suggests the estimation of a *total* effect of field of degree on earnings, without controlling for occupation, rather than the *partial* effect described above.

Accordingly, we implement a *sequential* decomposition framework, like that used in Altonji et al. (2012), to measure the gender pay gap and how it is affected by our explanatory variables of interest. We start with the baseline regression model

$$(1) \qquad y_{id} = \alpha_d^0 + \beta_d^0 W_{id} + \varepsilon_{id}$$

[11] If eventual field and occupation outcomes were predetermined on entry to college by a set of other factors, then it would be wrong to suspect such a dependence. However, the literature suggests that student experiences in college are critical for the choice of major, as they receive information about their own preferences and skills through coursework and extracurricular activities, and update their priors (e.g. Zafar 2011; Stinebrickner and Stinebrickner 2014; Kahn and Ginther 2017; Patnaik et al. 2020). Moreover, different fields confer different skills that are relevant for earnings outcomes (e.g. Arcidiacono 2004). Work by Calkins et al. (2022) finds that a large set of baseline characteristics, including *intent* to major in STEM as a freshman, explains only one quarter of observed variation in the decision to complete a STEM degree.

where *y* is an earnings outcome, *W* is a binary indicator taking the value 1 if individual *i* is a woman, and *d* is the given stratum (e.g. degree level).[12] When *y* corresponds to earnings in levels, the baseline gender pay gap estimate is $\widehat{\beta_d^0}/\widehat{\alpha_d^0}$, which is the earnings difference between the average woman and average man expressed as a percentage of his earnings. When *y* corresponds to earnings in logs, the baseline estimate is $\widehat{\beta_d^0}$, which has the standard log-point interpretation.

We then sequentially add controls for 1) field of degree, 2) occupation and industry, and 3) labor supply, re-estimating the model each time and tracking how the coefficient on *W* changes. For example, to measure the contribution of field of degree to the gender pay gap, we estimate

$$(2) \qquad y_{id} = \alpha_d^1 + \beta_d^1 W_{id} + \boldsymbol{\gamma_d}' \boldsymbol{F_{id}} + \varepsilon_{id}$$

where $\boldsymbol{F}$ denotes the suite of two-digit CIP code fixed effects. The amount of the gender gap explained by gender differences in field of degree is given by $\widehat{\beta_d^0} - \widehat{\beta_d^1}$, while the remaining gap is $\widehat{\beta_d^1}$. Subsequent iterations yield the amount of the gap explained by occupation and industry differences conditional on field choices; the amount explained by labor supply differences conditional on occupation, industry, and field; and the residual portion left unexplained.

While the ordering of field-of-degree choices before occupation and labor supply is "natural", it is worth elaborating on our decision to control for occupation ahead of labor supply. One might argue that individuals set their occupation and labor supply choices simultaneously, or that labor supply preferences are independent of occupation-specific preferences and skills. In this case, one should not expect an intervention that causes a change in graduate *i*'s occupation choice to have spillover effects on *i*'s labor supply. Such a process is more consistent with first controlling for field of degree, and then estimating a decomposition on the residual gap that controls simultaneously for occupation and labor supply. Our decision to proceed sequentially is consistent with assumptions that labor supply is one of a correlated set of considerations that drive occupation choice. It also allows our study to engage with the popular hypothesis that the sorting of men and women into higher- versus lower-hours positions is an important mechanism of the gender gap within skilled professional occupations (Goldin 2014).

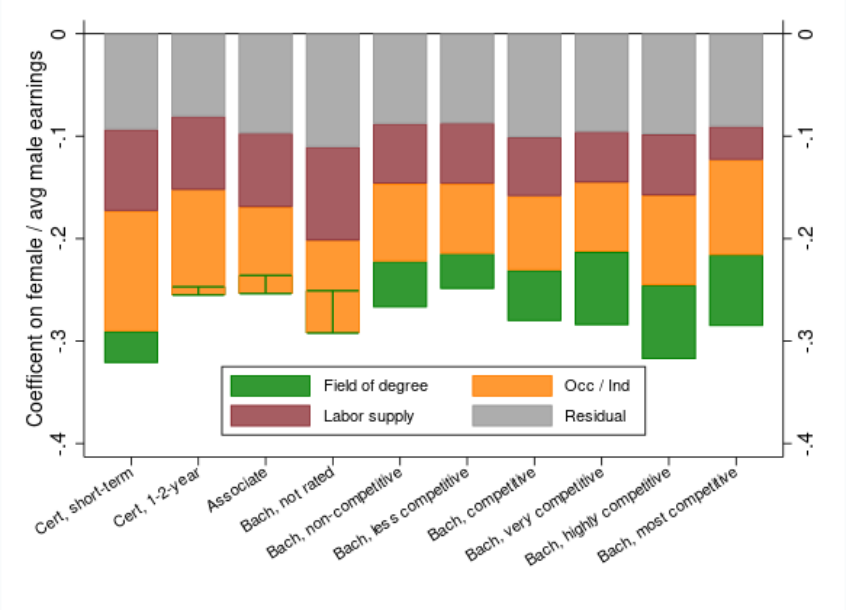
One drawback of our sequential approach is that, by not considering the covariance structure of the explanatory variables, it does not allow for extremely precise inference. Specifically, because the contribution of explanatory factor *n*+1 is determined by $\widehat{\beta_d^n} - \widehat{\beta_d^{n+1}}$, and each $\widehat{\beta_d^n}$ (*n*=0,1,2,3) coefficient is estimated independently, the standard error of this contribution must be inferred as the square root of the sum of the variances of the two coefficient estimates. In some cases, we find economically meaningful heterogeneity in the gender gap and its mechanisms across degree levels that is not statistically significant. We discuss these cases as they arise.

[12] The baseline regression model controls for a suite of age, survey year, and graduation year fixed effects as well, which are absorbed into the constant term when the decompositions are computed.

IV. Main Gender Pay Gap Estimates

Figure 1 presents gender pay gap decompositions for each of the ten levels of degree. The total height of each bar corresponds to the baseline gender pay gap estimate (i.e. after controlling for age, graduation year, and survey year effects). Green is the portion of the gap explained by gender differences in field of degree. Orange is the portion of the remaining gap explained by occupation and industry differences. Maroon is the portion of the remaining gap explained by labor supply differences. Gray represents the residual portion. In a few cases, controlling for field of degree widens, rather than narrows, the gender pay gap. These cases are captured by green whiskers as opposed to opaque bars.

Panel A. Total Earnings



Panel B. Log Wage and Salary Earnings

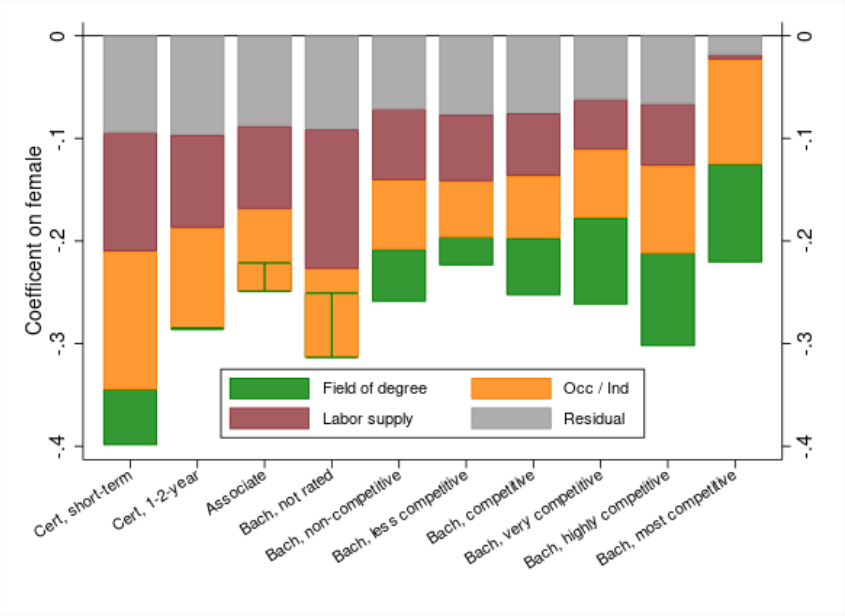


FIGURE 1. GENDER PAY GAP DECOMPOSITIONS BY TYPE OF DEGREE, FULL SAMPLE

Source: PSEO transcript records linked to the 2005-2019 ACS and the 2004 Barron's Guide.
Note: See Section II.A for sample information. "Cert" stands for certificate degrees and "Bach" for bachelor's degrees. "Occ / Ind" stands for occupation and industry effects. Total length of each bar corresponds to the gender pay gap after controlling for age, degree year, and survey year effects. Green is the portion of the gap explained by adding 2-digit CIP code fixed effects to the pay gap regression. Orange is the portion of the remaining gap explained by adding 315 occupation and 19 industry fixed effects. Maroon is the portion of the remaining gap explained by adding 48 fixed effects describing different levels of labor supply (6 weeks worked bins interacted with 8 weekly hours bins). Gray represents the residual portion. In a few cases, controlling for field of degree widens, rather than narrows, the gender pay gap. These cases are captured by green whiskers as opposed to opaque bars.

Although recent literature has focused on the very top of the human capital distribution, Panel A shows a raw gender pay gap that is consistent across degree types. Panel B, which considers log wage-and-salary earnings, shows a raw gap that generally *falls* as degree level rises. Both panels show that the remaining gap after controlling for field and occupation sorting (i.e. the gray + maroon portion) is smaller at higher degree levels. We also find that field-of-degree sorting contributes importantly to the gender pay gap among those graduating from top bachelor's programs, but has a smaller effect on the pay gap among less-selective bachelor's, certificate, and associate's degree holders. In contrast, labor supply differences determine the pay gap to a larger extent at these degree levels.

TABLE 4. GENDER PAY GAP DECOMPOSITIONS BY BINNED DEGREE LEVEL

	Non-Bachelor		Bachelor			
	Certificate	Associate	Not Selective	Selective	Very Selective	Most Selective
Panel A. Total Earnings, Full Sample						
Total gap	-0.288	**-0.236**	-0.257	-0.280	-0.300	-0.284
	(0.008)	(0.004)	(0.005)	(0.004)	(0.005)	(0.016)
Field of degree	**-0.011**	**0.018**	**-0.024**	-0.050	-0.072	-0.070
	(0.012)	(0.006)	(0.007)	(0.006)	(0.007)	(0.023)
Occupation	-0.111	-0.085	-0.077	-0.072	-0.077	-0.092
	(0.014)	(0.006)	(0.007)	(0.006)	(0.007)	(0.023)
Labor supply	**-0.076**	**-0.072**	-0.065	-0.057	-0.054	-0.032
	(0.013)	(0.006)	(0.007)	(0.006)	(0.007)	(0.021)
Residual	-0.090	-0.097	-0.091	-0.101	-0.097	-0.090
	(0.009)	(0.004)	(0.005)	(0.004)	(0.004)	(0.015)
N	54,500	192,000	90,500	150,000	130,000	12,500
Panel B. Log Wage and Salary Earnings, Full Sample						
Total gap	**-0.347**	-0.221	-0.243	-0.253	**-0.279**	-0.221
	(0.009)	(0.005)	(0.006)	(0.005)	(0.005)	(0.019)
Field of degree	**-0.027**	**0.028**	**-0.019**	-0.056	-0.087	-0.097
	(0.015)	(0.007)	(0.009)	(0.007)	(0.008)	(0.027)
Occupation	-0.119	-0.080	-0.068	-0.060	-0.075	-0.102
	(0.016)	(0.007)	(0.010)	(0.007)	(0.008)	(0.027)
Labor supply	**-0.105**	**-0.081**	**-0.079**	**-0.061**	**-0.053**	-0.005
	(0.014)	(0.006)	(0.008)	(0.006)	(0.007)	(0.023)
Residual	**-0.097**	**-0.088**	**-0.077**	**-0.076**	**-0.064**	-0.017
	(0.008)	(0.003)	(0.005)	(0.004)	(0.004)	(0.013)
N	46,000	168,000	81,500	135,000	118,000	11,000

Source: PSEO transcript records linked to the 2005-2019 ACS and the 2004 Barron's Guide.
Note: Ordinary least squares models with robust standard errors appearing below coefficient estimates in parentheses. **Bold** point estimates are significantly different from the estimate for Most Selective bachelor's graduates at the 10 percent level or less.

Table 4 presents decomposition estimates and standard errors to provide precise numbers behind Figure 1. For ease of exposition, we collapse the ten degree levels into six categories. We pool the two types of certificate degrees together, collapse the six bachelor's categories that are not "Most Competitive" down to three categories, and preserve the Most Competitive bachelor's category—which contains around two percent of the sample—given the prior literature's focus on the top of the human capital distribution. For the five other categories, we highlight in bold any point estimate that differs statistically from the corresponding estimate for the top category at the 10 percent significance level or less.

This table solidifies the graphical results. First, for total earnings, the pay gap is quite constant across degree levels; whereas for log wage and salary earnings, the pay gap is noticeably larger at the certificate level than it is at the Most Competitive bachelor's level. Second, field of degree sorting explains 25-43 percent of the gap among Most Competitive bachelor's graduates (depending on earnings measure) but is a less important determinant of the gap at the sub-

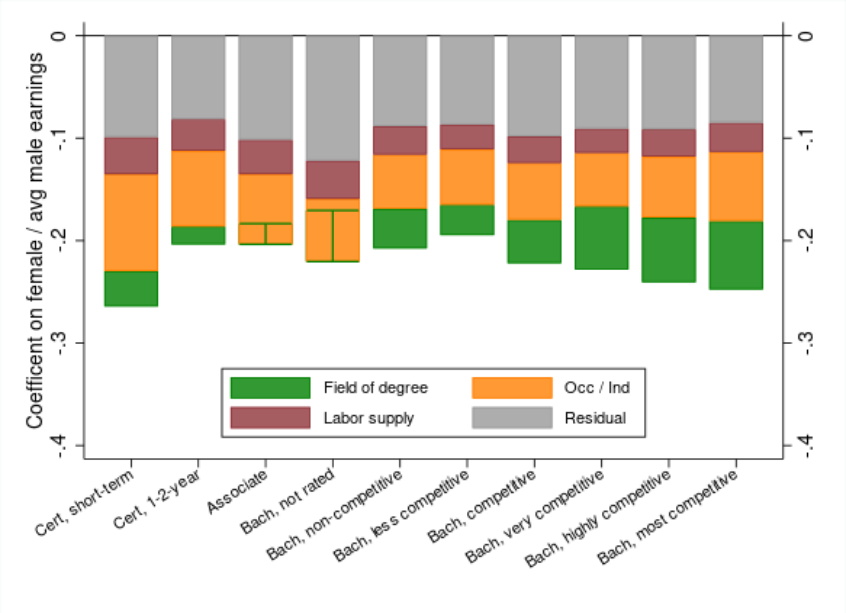
baccalaureate and less-selective bachelor's levels.[13] Third, occupation sorting conditional on field-of-degree sorting is an important determinant of the gender pay gap, explaining 30-40 percent of the gap at all degree levels. Fourth, occupation-conditional labor supply differences are an important determinant of the gap at sub-baccalaureate degree levels, explaining approximately 30 percent of the gap. In contrast, labor supply is a trivial determinant of the gap among Most Competitive bachelor's graduates. Fifth, approximately one-third of the *total* earnings gap remains unexplained at all degree levels; however, the unexplained portion of the *log wage-and-salary* earnings gap falls as degree level rises. Only 7.7 percent of the log wage-and-salary earnings gap among Most Competitive bachelor's graduates is unexplained by field of degree, occupation, and labor supply effects.

A. Estimates for Full-Year Workers

In addition to considering the full population, the gender gap literature often takes interest in the subset of full-year-employed workers: a group with "fairly similar levels of labor-market commitment" (Blau and Kahn 2017, p. 792). Figure 2 presents gender pay gap decompositions by degree level for this group. The results are akin what was shown above, with three slight exceptions. First, the total earnings gap is noticeably smaller at all degree levels. Second, the labor supply channel is estimated to be small at all degree levels, regardless of earnings measure. This suggests that the larger labor supply contributions found at less-selective-bachelor's and sub-baccalaureate degree levels in the full sample are due to gender differences in the extensive margin of labor supply. Third, occupation effects remain substantial at all degree levels, but their magnitudes are slightly smaller. In any case, both panels still show that after controlling for field and occupation effects, the remaining gender pay gap falls as degree level rises (and the raw gap is relatively constant across degree levels).

[13] In results not presented here, we examine how our estimates change if we use four-digit as opposed to two-digit CIP codes. While residual gender pay gaps, as well as the portions explained by labor supply effects, remain remarkably similar when we use four-digit CIP codes, the portion of the gender pay gap explained by field of degree increases moderately for certificate degree holders, with an offsetting decrease in the portion of the gap explained by occupation sorting (and a smaller offsetting decrease in labor supply and residual effects). Field of degree (labor supply) still affects the gender gap substantially more (less) for Most Selective bachelor's programs than for certificate programs, although the field-of-degree comparison loses statistical significance. (All other significant statistical comparisons bolded in Table 4 are preserved in the alternate specification.) We note that at the certificate level, four-digit CIP codes correspond closely to occupation choices, so the distinction between field of degree and occupation choices is somewhat murky—a fact which appears to be reflected in the offsetting changes to the explanatory power of field of degree versus occupation.

Panel A. Total Earnings



Panel B. Log Wage and Salary Earnings

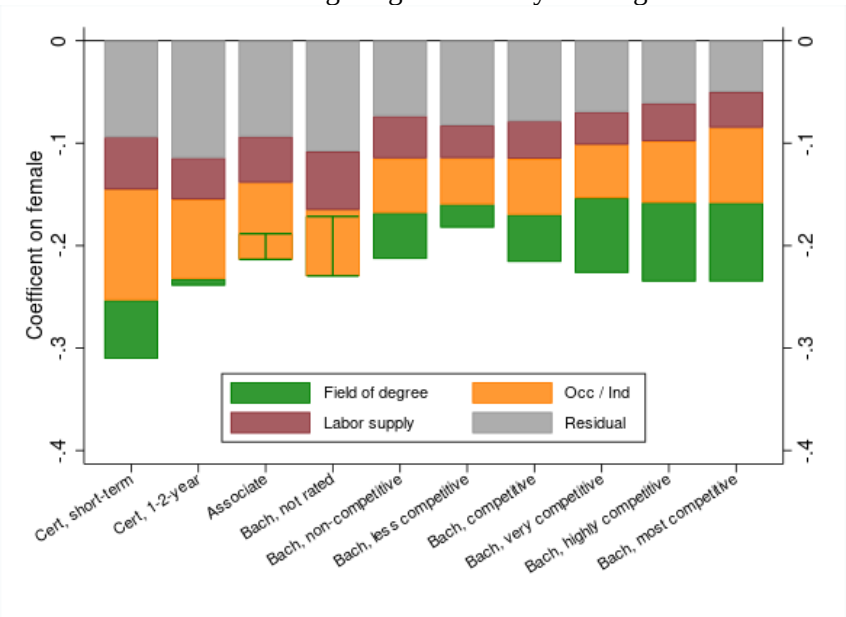


FIGURE 2. GENDER PAY GAP DECOMPOSITIONS BY TYPE OF DEGREE, YEAR-ROUND WORKERS

Source: PSEO transcript records linked to the 2005-2019 ACS and the 2004 Barron's Guide.
Note: See Section II.A for sample information. Sample restricted to individuals employed for 50-52 weeks during the survey year. "Cert" stands for certificate degrees and "Bach" for bachelor's degrees. "Occ / Ind" stands for occupation and industry effects. See notes to Figure 1 for further information on the bar graphs.

B. Estimates by Time since Graduation

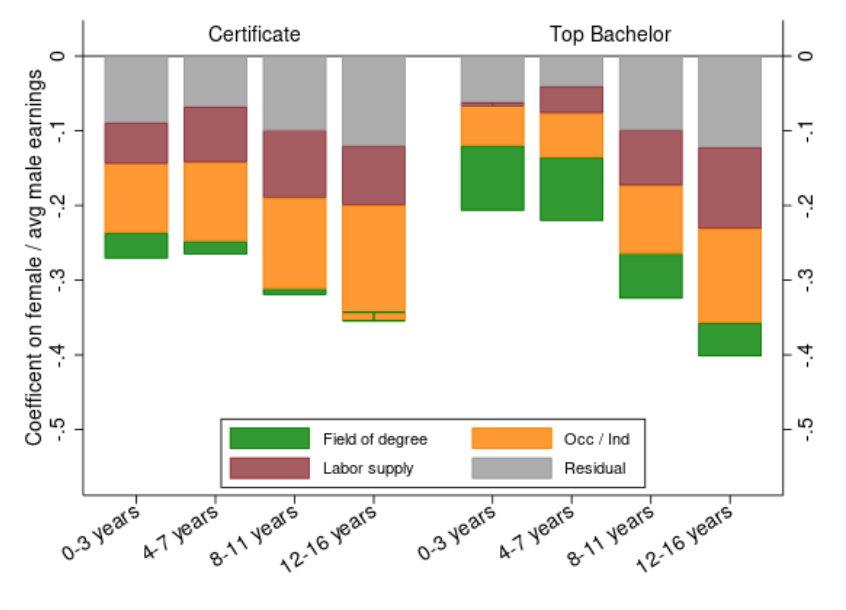
Next, we consider how the gender pay gap and its proximate mechanisms vary with time since graduation. As mentioned in the Introduction, one strand of the gender pay gap literature examines pay gap evolution early in career as individuals transition into the labor market, search for and possibly move to jobs with higher match quality, and form families. Our dataset allows us to update this literature by focusing on recent graduating cohorts, but also to expand it by investigating how the evolution of the pay gap over time varies with degree level. To keep the number of comparisons manageable and to increase statistical power, for the remainder of this section we focus on two different degree levels: certificate (formed by pooling all certificate degree earners together), and "Top" bachelor's (formed by pooling Highly and Most Competitive bachelor's degree earners together).

Panel A of Figure 3 shows, consistent with the prior literature, that the gender pay gap rises substantially with time since graduation.[14] However, at both degree levels, this increase does not begin until 7 years after graduation, after which there is a marked increase between years 7 and 16. For both degree levels, this increase in the overall gap is driven by an increase in occupation effects and in unexplained channels. Additionally, for the Top bachelor's group, there is a marked increase in labor supply effects between years 0-3 and 12-16, from virtually 0 percentage points to over 10 percentage points. There is also a moderate decline in field of degree effects, which initially explain one-half of the gap, as downstream channels become more important determinants of gender pay differences. Panel A also shows that the overall constancy of the gender pay gap across degree levels (Figure 1) masks different time patterns: the gap at 0-3 years after graduation is slightly larger for certificate graduates, while the gap at 12-16 years is slightly larger for Top bachelor's graduates. The increase in the total earnings gap between years 4-7 and 12-16 amounts to 18.1 percentage points for Top bachelor's graduates, but only 7.7 percentage points for certificate graduates, and this difference is statistically significant.

Panel B, which considers log wage-and-salary earnings, illustrates similar patterns, with two notable differences. First, the pay gap at each point in time is larger for certificate graduates than for Top bachelor's graduates—consistent with our earlier finding that the log wage-and-salary earnings gap is largest at the certificate level (Table 4). Second, labor supply effects become more important with time for certificate graduates as well as Top bachelor's graduates, although the time gradient remains stronger for the latter. Overall, the increase in the log wage-and-salary earnings gap between years 4-7 and 12-16 amounts to 18.6 log points for Top bachelor's graduates and 13.1 log points for certificate graduates (although this difference is not statistically significant).

[14] Because we do not observe the full 16 years of earnings trajectories for all graduating cohorts we consider, underlying cohort membership is not held constant across each time-since-graduation cell. Appendix Figure A1 tests for cohort effects by plotting the main gender pay gap decompositions for each of three different cohort groups (2003-05, 2006-08, and 2009-13) during the range of years that are observed for all three groups (0-6 years following graduation). The plots show similar decompositions, at both certificate and Top bachelor's levels, for all three cohort groups. This suggests that the dynamics found in Figure 3 reflect time effects rather than cohort effects.

Panel A. Total Earnings



Panel B. Log Wage and Salary Earnings

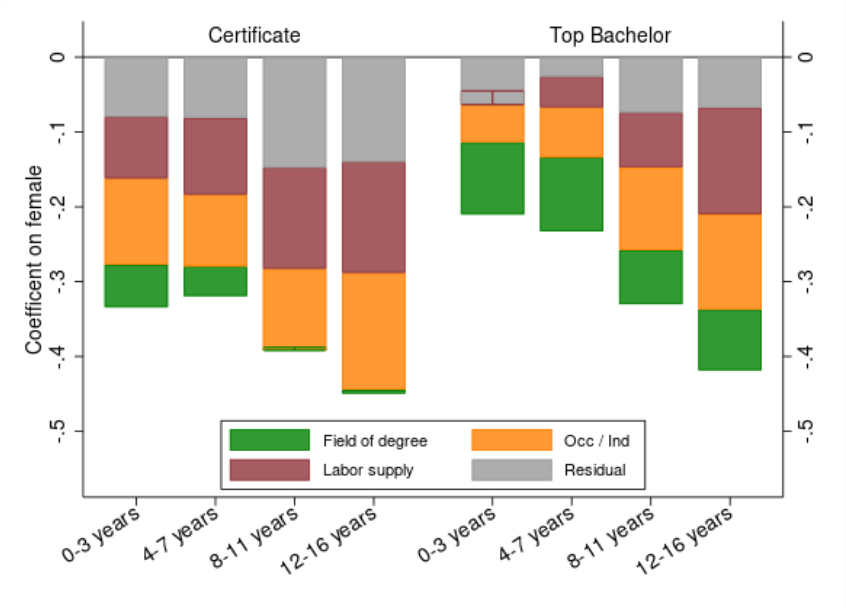


FIGURE 3. GENDER PAY GAP DECOMPOSITIONS BY TIME SINCE DEGREE

Source: PSEO transcript records linked to the 2005-2019 ACS and the 2004 Barron's Guide.
Note: See Section II.A for sample information. "Top Bachelor" refers to graduates from Most Competitive or Highly Competitive institutions, as rated by Barron's. See notes to Figure 1 for further information on the bar graphs.

C. The Role of Children

Considering recent literature showing strong effects of parenthood on the gender pay gap (e.g. Juhn and McCue 2017), we utilize the CHCK subsample to perform gender pay gap decompositions for those without and those with children. The difference between these two provides a descriptive estimate of how children influence the gender pay gap and each of its mechanisms. This is akin to interacting the main specifications with a control for children and reporting the coefficient estimate on that control.

TABLE 5. GENDER PAY GAP DECOMPOSITIONS BY PARENTAL STATUS

	Certificate Grads			*Top Bachelor Grads*		
	With Children	Without Children	Difference	With Children	Without Children	Difference
Panel A. Total Earnings						
Total gap	-0.441	-0.109	**-0.332**	-0.530	-0.199	**-0.331**
	(0.019)	(0.028)	(0.034)	(0.018)	(0.014)	(0.023)
Field of degree	-0.025	-0.007	-0.018	-0.052	-0.059	0.007
	(0.032)	(0.044)	(0.054)	(0.027)	(0.021)	(0.034)
Occupation	-0.115	-0.089	-0.026	-0.163	-0.061	**-0.102**
	(0.038)	(0.048)	(0.061)	(0.028)	(0.021)	(0.035)
Labor supply	-0.125	-0.020	**-0.105**	-0.183	-0.014	**-0.169**
	(0.039)	(0.048)	(0.062)	(0.029)	(0.020)	(0.035)
Residual	-0.176	0.007	**-0.183**	-0.132	-0.065	**-0.067**
	(0.027)	(0.033)	(0.043)	(0.020)	(0.014)	(0.024)
N	6,000	4,400		7,200	13,100	
Panel B. Log Wage and Salary Earnings						
Total gap	-0.580	-0.162	**-0.418**	-0.645	-0.167	**-0.478**
	(0.024)	(0.029)	(0.038)	(0.024)	(0.015)	(0.028)
Field of degree	-0.041	-0.005	-0.036	-0.084	-0.074	-0.010
	(0.042)	(0.045)	(0.062)	(0.036)	(0.022)	(0.042)
Occupation	-0.119	-0.086	-0.033	-0.188	-0.063	**-0.125**
	(0.050)	(0.050)	(0.071)	(0.036)	(0.022)	(0.042)
Labor supply	-0.206	-0.041	**-0.165**	-0.253	-0.008	**-0.245**
	(0.045)	(0.045)	(0.064)	(0.032)	(0.019)	(0.037)
Residual	-0.214	-0.030	**-0.184**	-0.120	-0.022	**-0.098**
	(0.026)	(0.027)	(0.037)	(0.019)	(0.012)	(0.022)
N	5,100	3,900		6,200	12,400	

Source: PSEO transcript records linked to the 2005-2019 ACS, the 2004 Barron's Guide, and the Census Household Composition Key (CHCK).
Note: "Top Bachelor" refers to graduates from Most Competitive or Highly Competitive institutions, as rated by Barron's. Ordinary least squares models with robust standard errors appearing below coefficient estimates in parentheses. Estimates in **bold** denote statistically significant effects of children on the gender pay gap or its associated component at the 10 percent level or less.

Table 5 presents the results. Panel A shows that the total earnings gap is around 33 percentage points higher for individuals with children; Panel B shows that the log wage-and-salary earnings gap is around 45 log points higher for individuals with children. The small CHCK subsample sizes lead to a lack of precision in some of the estimates, but we still find that children significantly widen the total earnings gap through labor supply and residual channels for certificate graduates, and through occupation, labor supply and residual channels for Top bachelor's graduates.

(Moreover, the difference in the residual effects of children across degree levels is statistically significant.) We find a similar pattern of effects for the log wage-and-salary earnings gap. Overall, these results suggest that children widen gender pay gaps through several different channels, and these channels differ according to degree level. In particular, the greater effect of children on residual channels for certificate graduates could be the result of less access to employer-provided benefits, less access to formal or informal support networks, or more exposure to various forms of discrimination for these workers (see Cortés and Pan 2020 for further discussion). It could also be the result of greater children-induced work interruptions and losses of human capital for mothers (or corresponding gains for fathers), although the finding of smaller children-induced labor supply effects casts doubt on this story. In contrast, children appear to lead to occupational downgrading for women (and upgrading for men) to a larger extent among Top bachelor's graduates.

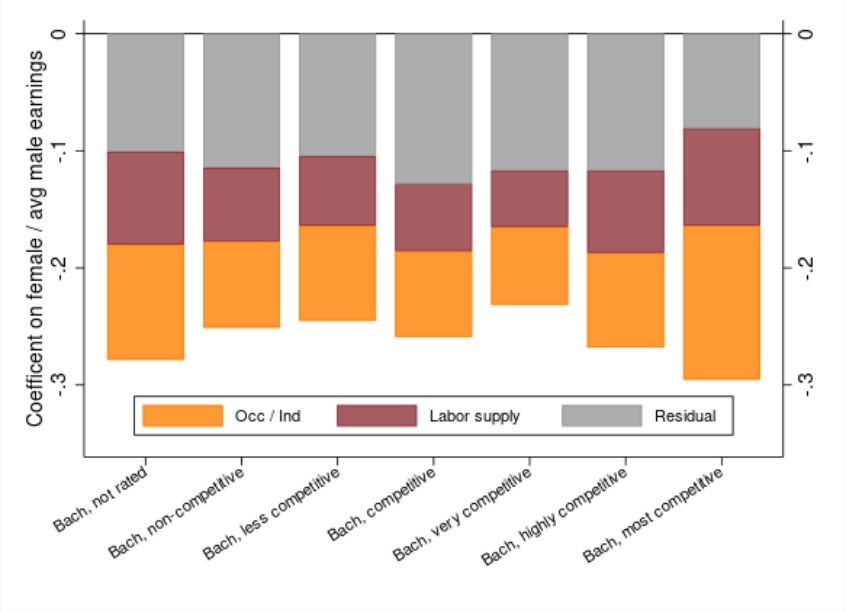
V. Pay Gap Estimates for Business Majors

In this section, we repeat some of the above analyses for the subsample of individuals who graduated with a bachelor's degree and majored in Business-related fields.[15] Business is the single largest CIP code in our sample, accounting for roughly 20 percent of all bachelor's graduates. Moreover, business-related occupations are often highlighted as having some of the largest gender pay gaps (see for example, Goldin 2014 and Bertrand, Goldin, and Katz 2010). Our sample affords us a unique opportunity to analyze the career trajectories of Business majors across different degree levels. We compare our findings, both general and Business-major-specific, to related studies in the next section.

Figure 4 repeats the main gender pay gap decompositions presented in Figure 1 for our subsample of Business majors with bachelor's degrees. Because all graduates in this subsample chose the same major, we omit field-of-degree effects from the model. We find a pattern of results that is quite consistent with the findings for the full sample. Specifically, the total earnings gap is constant across Barron's ratings, with some evidence that the residual earnings gap is smallest among Business graduates from Most Competitive institutions. Moreover, the log wage-and-salary earnings gap—both with and without controls for occupation and labor supply—falls as selectivity rises. At the Most Competitive level, the log wage-and-salary earnings gap is approximately 15 log points, and it is *entirely* explained by occupation effects. After controlling for occupation and labor supply effects, women earn nearly 4 log points *more* than comparable men, although this difference is not statistically significant.

[15] Business fields have their own 2-digit CIP code of 52. We also pulled Economics majors out of the Social Sciences CIP code of 45 and pooled them with the Business majors.

Panel A. Total Earnings



Panel B. Log Wage and Salary Earnings

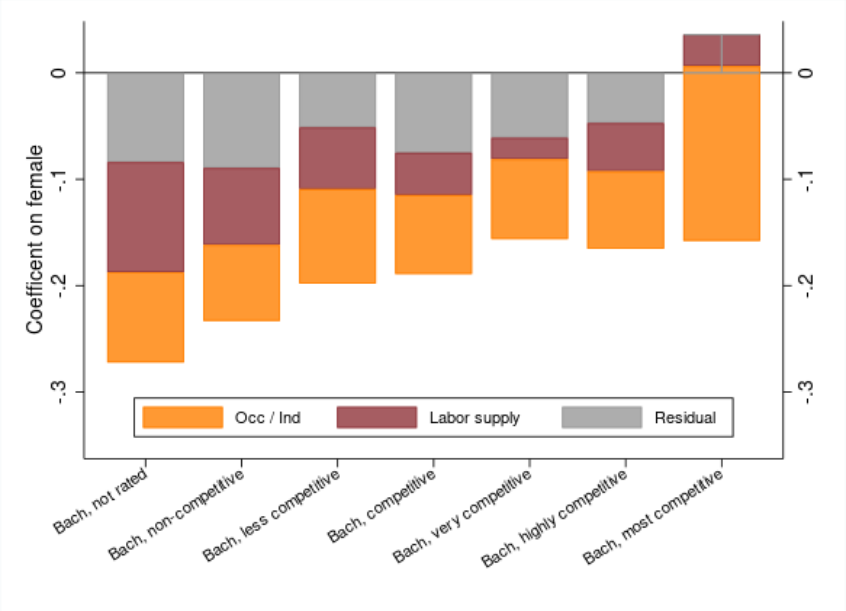


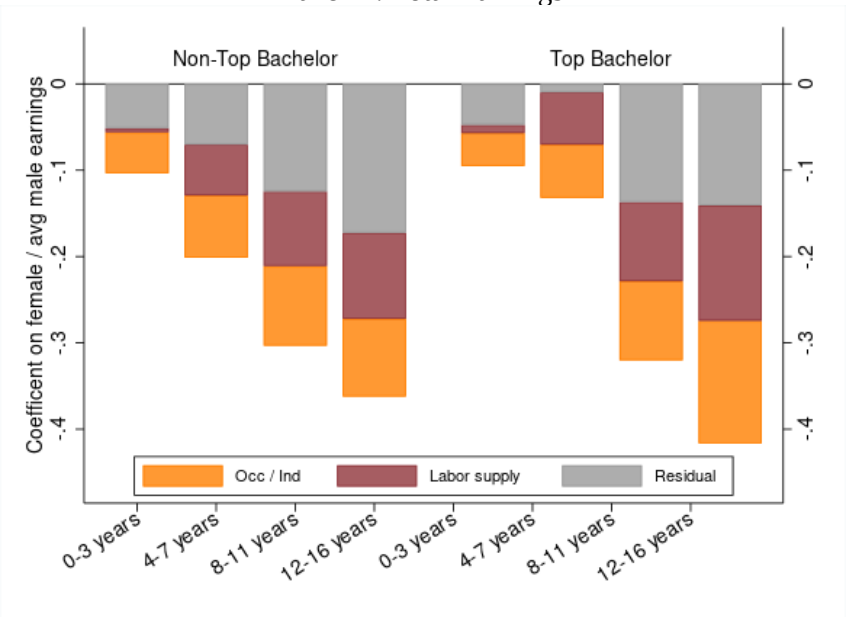
FIGURE 4. GENDER PAY GAP DECOMPOSITIONS FOR BUSINESS MAJORS BY DEGREE LEVEL

Source: PSEO transcript records linked to the 2005-2019 ACS and the 2004 Barron's Guide.
Note: See Section II.A for sample information. Sample restricted to Business and Economics graduates. "Bach" refers to bachelor's degree. See notes to Figure 1 for further information on the bar graphs.

Next, we repeated the analysis of Section IV.B to analyze the evolution of the gender pay gap with time since graduation. As before, we pooled the top two Barron's categories together into a Top bachelor's stratum. We compared this stratum to a stratum containing the lower five categories (denoted as "Non-Top" bachelor's). Figure 5 presents the results. According to Panel A, there is a small but statistically significant total earnings gap of around 10 percent in years 0-3 following graduation. This gap increases continuously for Non-Top bachelor's degree earners, as labor supply, residual, and (to a lesser extent) occupation effects all increase in magnitude. After 12-16 years, the total gap rises to approximately 36 percent. We observe a similar pattern for Top bachelor's degree earners, although the increase in the gap is more punctuated between years 4-7 and 8-11. All three channels (occupation, labor supply, and residual) increase with time to a slightly larger extent for Top bachelor's graduates, such that by 12-16 years after graduation, the total gap rises to approximately 42 percent.

Panel B illustrates similar dynamics for the log wage-and-salary earnings gap, except that the initial gap upon graduation is estimated to be even smaller. For Non-Top graduates, the gap in years 0-3 following graduation is only 6.9 log points; for Top graduates, the initial gap is only 3.9 log points, and the estimate is statistically insignificant. By 12-16 years after graduation, the total gap rises to approximately 36.5 log points for both degree levels.

Panel A. Total Earnings



Panel B. Log Wage and Salary Earnings

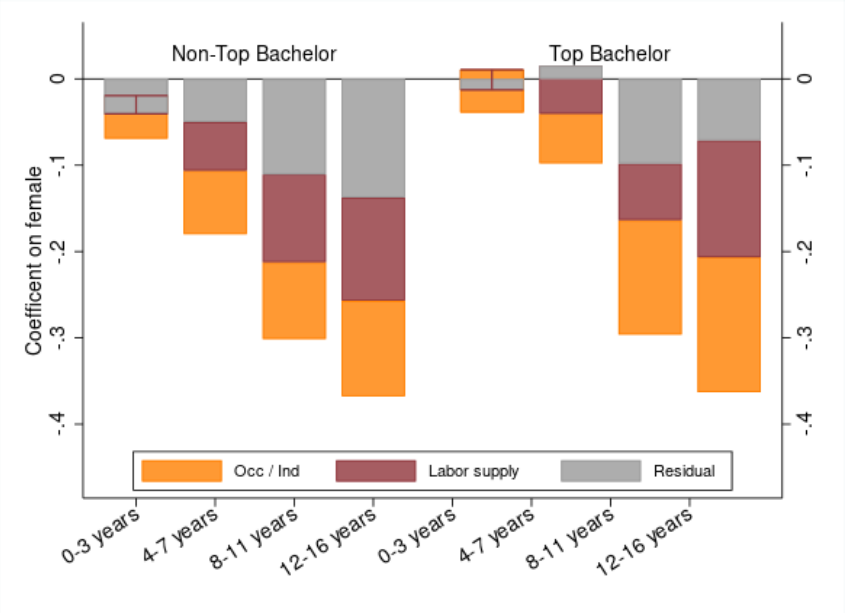


FIGURE 5. GENDER PAY GAP DECOMPOSITIONS FOR BUSINESS MAJORS BY DEGREE LEVEL

Source: PSEO transcript records linked to the 2005-2019 ACS and the 2004 Barron's Guide.
Note: See Section II.A for sample information. Sample restricted to Business and Economics graduates. "Bach" refers to bachelor's degree. See notes to Figure 1 for further information on the bar graphs.

VI. Comparison to Related Literature

Our analyses relate to several other studies in the literature that focus on early-career gender pay gaps. It is instructive to compare our results to those from these studies—despite some differences in sample membership and gender gap measurement—as a review of the literature and to learn about how the gap has changed in the modern cohorts we study. We classify comparisons into two different types.

First, some prior studies have estimated the contributions of field-of-degree sorting to the gender pay gap. Brown and Corcoran (1997) found, in a sample of respondents to the National Longitudinal Study of the High School Class of 1972, who were surveyed in 1986 when they were in their early 30s, that field of bachelor's degree explained 7.8 log points (37.7 percent) of a total 20.7 log-point gap. We compared our sample to theirs by repeating our sequential decomposition analysis on bachelor's graduates, observed 7-10 years since graduation, with log hourly wages as the outcome. We found that field-of-degree effects explained 4.9 log points (27.5 percent) of a total 17.8 log-point gap. This suggests less—but still substantial—gender segregation of field choice in today's U.S. cohorts as compared to 30 years ago.[16] Francesconi and Parey (2018—hereinafter FP) examined the gender pay gap among 1989-2009 German university graduates in the first two years following graduation. They found that field of study effects explained 9.4 log points (49.2 percent) of the total 19.1 log-point gap in monthly earnings among fulltime workers.

[16] See Bailey and DiPrete (2016) for further discussion.

Our comparable estimate for bachelor's graduates in our sample is a field-of-degree contribution of 4.7 log points (34.3 percent) to a total 13.7 log-point gap.

Most recently, Sloane et al. (2021) examined the contribution of field-of-degree and occupational sorting to the full-time-employed hourly wage gap among college graduates sampled by the ACS in 2014-17. In their sample of 1978-87 birth cohorts (who would have graduated college just a few years earlier than our cohorts on average), field and occupation controls jointly explained 9.0 log points, or 58.1 percent, of the observed 15.5 log-point gap.[17] This estimate exceeds our estimate for less-competitive bachelor's programs but is similar to our findings for more-competitive programs.

Second, a host of studies have examined the evolution of the early-career gender pay gap, often in specialized settings (e.g. Wood et al. 1993). Bertrand, Goldin and Katz (2010—hereinafter BGK) previously found a large gender pay gap that grew considerably with time since graduation in a sample of University of Chicago MBA grads. In comparing our results to BGK's, it is important to recognize that our samples of Business graduates differ from BGK's sample. MBA graduates are positively selected relative to the average bachelor's degree holder (Nevill and Chen 2007), but the BGK sample is even more positively selected: Booth Business School consistently ranks among the top 5 MBA programs in the country. Moreover, MBA graduates tend to come from a wide array of undergraduate fields, and business undergraduates are somewhat less likely to attain graduate education compared to other bachelor's degree holders (Nevill and Chen 2007; Altonji et al. 2016). Our samples of Business graduates isolate a different, and potentially more-representative, subset of individuals trained for business-related occupations.

Table 6 compares main gender pay gap estimates between our samples of business graduates and BGK's.[18] Note that time since graduation refers to years since earning a bachelor's degree in our sample, and years since earning an MBA in BGK's. Comparing the results simply by years since graduation, log earnings gaps are substantially smaller in our samples relative to BGK's. For example, we find pay gaps of 6.9 and 3.9 log points for Non-Top and Top bachelor's graduates 0-3 years after graduation, respectively, and these gaps grow to 30.1 and 29.6 log points at 8-11 years after graduation. BGK estimated an 18.9 log point gap at 0-3 years and a 44.7 log point gap at 8-11 years after graduation—roughly 15 log points larger than our estimates. If we instead compare the MBA sample to similarly-aged bachelor's samples (i.e., comparing MBAs 0-3 years after graduation to bachelor's graduates 4-7 years after graduation and so on), we find less-disparate results. The very early-career gap does appear to be smaller for Top bachelor's graduates at 9.8 log points, compared to 18.9 log points for the BGK sample. However, Non-Top bachelor's graduates face a similar pay gap very early in career (18.0 log points). Moreover, while the gaps at 12-16 years postgraduation in our samples are still smaller than the gap at 8-11 years postgraduation in BGK's sample, the differences in gap estimates are less than 8.5 log points (36.8 and 36.3 log points in our samples compared to 44.7 log points in BGK's).

[17] Since Sloane et al. (2021) introduced field of degree controls *after* occupation controls in their empirical model, it is impossible to compare their separate estimates of field and occupation effects to ours. They also controlled for field of degree and occupation using a slightly different methodology that emphasizes male returns to those variables, rather than the gender-neutral average returns implied by our fixed effects.

[18] Appendix section B describes how we aligned BGK's estimates with ours.

Overall, we take this comparison as evidence that today's typical Business graduates face less-severe gender inequality than that reflected in BGK's sample, particularly at early career stages. However, our finding of a stark increase in the pay gap with time since graduation matches BGK's, and suggests that the family-related dynamics they emphasize remain relevant for the broader, contemporary population of Business graduates.

TABLE 6. COMPARISON TO BERTRAND, GOLDIN AND KATZ (2010)

Time since graduation	Log earnings gap				
	BGK	Our study		Difference w/ BGK	
		Non-Top	Top	Non-Top	Top
0-3 years	-0.189	-0.069	-0.039	-0.120	-0.144
4-7 years	-0.257	-0.180	-0.098	-0.077	-0.167
8-11 years	-0.447	-0.301	-0.296	-0.146	-0.146
12-16 years	--	-0.368	-0.363	--	--

Source: Authors' calculations based on published estimates from Bertrand et al. (2010); PSEO transcript records linked to the 2005-2019 ACS and the 2004 Barron's Guide.
Note: See Appendix for details on how we achieved an apples-to-apples comparison between our results and those reported by Bertrand et al. (2010).

In a related study of the early-career gender pay gap, FP reported a gap among full-time-working Business graduates of 11 log points in years 1-2 following graduation. We developed a comparable sample[19] and estimated a gap of 6.7 log points. In their full sample, FP reported a pay gap of 19.1 log points, while the estimate in our comparable sample is 13.7 log points. Finally, Manning and Swaffield (2008) studied the early-career evolution of the gender pay gap in a more general setting: a nationally representative (i.e. including college and noncollege-educated individuals) panel of British households observed during 1991-2002. They estimated an hourly wage gap of 14.4 log points at five years of potential experience and 24.8 log points at ten years. The corresponding estimates in our sample are 10.4 and 18.6 log points, respectively. These comparisons continue to suggest lower early-career gender inequality among contemporary American postsecondary graduates than has been reported in the literature.

VII. Conclusion

In this paper, we leverage a unique dataset of postsecondary transcript records linked to household survey data to provide a novel examination of the gender pay gap across the distribution of pre-labor-market human capital. This analysis expands a long literature on gender inequality that has either i) provided population estimates of the gender pay gap along with the contributions of human-capital variables; or ii) analyzed gender inequality in high-human-capital subpopulations. Our study is the first of which we are aware to interact these foci, breaking down a large sample of postsecondary degree earners into more narrowly defined human capital strata, and analyzing variation in the gender gap and its mechanisms across each stratum. We consider

[19] Francesconi and Parey (2018) do not report a definition of fulltime work. We used a definition of being employed for at least 40 weeks during the year and working at least 31 hours per week.

individuals that earned their postsecondary degrees in 2003-2013 and were observed up to 16 years following graduation; thus, our results shed light on contemporary early-career gender inequality.

Our approach yields a multitude of new findings. First, the gender pay gap is not highest for top postsecondary graduates. The total earnings gap (as a percentage of male earnings) is quite constant across the degree level distribution; the log wage-and-salary earnings gap *falls* as degree level rises. Second, field-of-degree sorting explains a substantial portion of the gap among Most Competitive bachelor's graduates but is a less important determinant of the gap at lower degree levels. Third, field-conditional occupation sorting is an important determinant of the gender pay gap at all levels. Fourth, occupation-conditional labor supply differences are an important determinant of the gap at sub-baccalaureate degree levels, but a fairly minor determinant at the top. Fifth, in the years immediately following graduation, the gap is smallest at the top and is mostly explained by field-of-degree sorting; after 12-16 years of labor-market exposure, the pay gap is similar at certificate and Top bachelor's levels. Sixth, the overall effects of children on the gender pay gap are similar at both levels, although they operate primarily through occupation and labor supply channels at the top, and through labor supply and residual channels at the bottom. Seventh, these findings generally hold in a sample of Business majors: the log wage-and-salary pay gap is smallest among Most Competitive bachelor's graduates and is entirely explained by occupation effects.

Comparisons of our findings to other related literature suggest that gender inequality has fallen among both bachelor's degree holders as a whole and among those with degrees from top institutions, relative to what has previously been reported. Relatedly, our findings indicate that within highly-skilled occupations, gender-based sorting into higher- versus lower-hours positions is less prevalent in recent cohorts than has previously been emphasized (e.g. Goldin 2014)—at least, at the relatively-early career stage we study. On the other hand, field-of-degree sorting remains an important determinant of the gender pay gap among recent bachelor's graduates, although slightly less important than in Brown and Corcoran's (1997) study. This suggests less, but still substantial, gender segregation in course-taking and completion that disadvantages women now as compared to 30 years ago. Moreover, our results on gender-segregation in field choice among graduates from top baccalaureate institutions mirrors Brown and Corcoran's results for the general population of bachelor's graduates.

Today's cohorts of postsecondary graduates continue to experience substantial gender inequality in labor market outcomes. We have shown that while the *magnitude* of this inequality does not vary much by level of credential, the *mechanisms* vary more substantially. This indicates the need for a mix of "global" and "human-capital-specific" policies to promote broad-based gender equality. Interventions aimed at equalizing the fields chosen by men and women—for example through interventions to the social and learning environments at postsecondary institutions—would significantly lessen the gender gap primarily for graduates from the most selective institutions and only relatively early in their careers. On the other hand, interventions to lessen gender-based occupational segregation—for example by strengthening alumni and other support networks in male-dominated occupations—could have a broad-based effect on gender inequality. Improving the provision of flexibility- and childcare-related benefits would help women maintain full-time labor supply and ensure more equitable treatment from employers upon the transition to parenthood. This would result in lower gender inequality particularly among less-

competitive bachelor's and sub-baccalaureate graduates—and would also lower the steep effects of time-since-graduation on the gender gap for top bachelor's graduates. Rather than simply *conditioning* on human capital, as has been the tradition of the literature, *interacting* gender and human capital strikes us as a fruitful approach to maintaining an applicable base of knowledge about the ever-present but ever-changing gender pay gap.

Appendix

A. Omitted Tables and Figures

TABLE A1. COMPARISON OF PSEO SAMPLE TO A NATIONALLY REPRESENTATIVE SAMPLE DRAWN FROM THE ACS

	PSEO sample	ACS sample
Share White non-Hispanic	0.71	0.63
Share Black non-Hispanic	0.09	0.12
Share Asian non-Hispanic	0.05	0.09
Avg total earnings, women	38,700	37,200
Avg total earnings, men	53,200	57,200
Share earning < $5000, women	0.17	0.20
Share earning < $5000, men	0.12	0.11

Source: PSEO transcript records linked to the 2005-2019 ACS; 2016 ACS 5-year public-use file.
Note: See Section II.A for PSEO sample information (left column). We compared main summary statistics from this sample to a nationally-representative sample drawn from the 2016 ACS 5-year public-use file (right column). This file contains observations from 2012-2016, with an average observation year of 2014, which matches our PSEO sample. We considered individuals with at least one year of completed postsecondary education and with 2-20 years of potential experience. This yielded a mean age of 31, which again matches our PSEO sample. Thus, the differences between the left and right columns are due to differences in sample membership other than average age.

TABLE A2. ACS EDUCATIONAL ATTAINMENT BY PSEO DEGREE LEVEL

PSEO degree level	Share with Bachelor degree	Share with advanced degree
Short-term Certificate	0.185	0.031
1-2-year Certificate	0.228	0.042
Associate	0.441	0.080
Unrated Bachelor	0.931	0.215
Non-Competitive	0.933	0.239
Less Competitive	0.929	0.223
Competitive	0.941	0.255
Very Competitive	0.952	0.308
Highly Competitive	0.952	0.338
Most Competitive	0.984	0.454

Source: PSEO transcript records linked to the 2005-2019 ACS and the 2004 Barron's Guide.
Note: See Section II.A for sample information. Sample restricted to individuals observed at least two years after earning their PSEO degree, and with non-imputed ACS educational attainment information. The bachelor's degree shares for PSEO bachelor's graduates (i.e. the last seven rows of the table) should be 100 percent, but are not due to mismeasurement of educational attainment in the ACS. See Foote and Warren (2020) for more detail.

Panel A. Total Earnings



Panel B. Log Wage and Salary Earnings

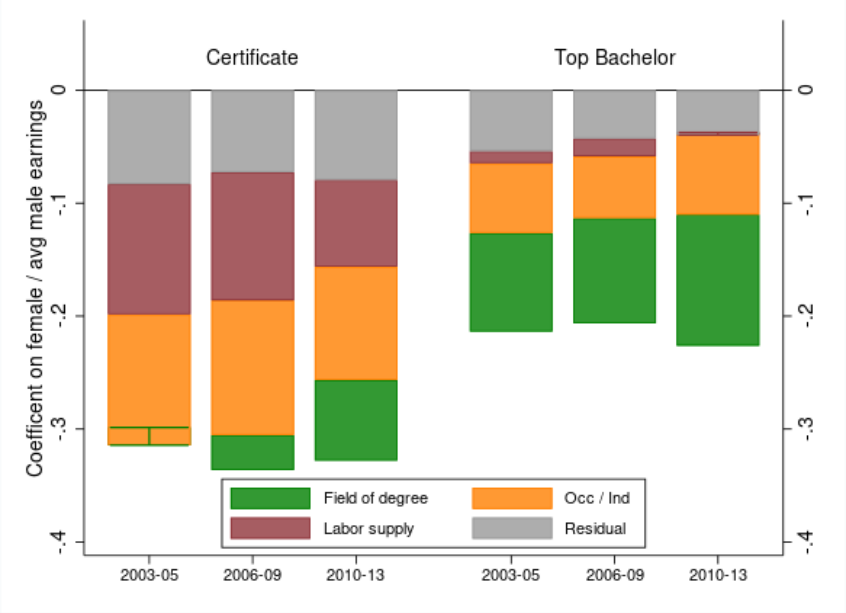


FIGURE A1. GENDER PAY GAP DECOMPOSITIONS BY GRADUATION COHORT, 0-6 YEARS FOLLOWING GRADUATION

Source: PSEO transcript records linked to the 2005-2019 ACS and the 2004 Barron's Guide.
Note: See Section II.A for sample information. Sample restricted to individuals observed during the common observation window across all cohorts: zero to six years following graduation. See notes to Figure 1 for further information on the bar graphs.

B. Comparing our Estimates to Those Reported by Bertrand et al. (2010—BGK)

We aligned our time periods with BGK's estimates (reported in their Table 2) as follows. To align our 0-3-year estimate with their estimates for years 0, 1, and 3, we applied weights of 0.25, 0.375, and 0.375, respectively, to their estimates. To align our 4-7-year estimates with their estimates for years 3, 6, and 9, we constructed a pseudo-year-4 estimate by applying weights of (0.67,0.33) to their year-3 and year-6 estimates. We constructed a pseudo-year-5 estimate by applying weights of (0.33,0.67) to their year-3 and year-6 estimates. We constructed a pseudo-year-7 estimate by applying weights of (0.67,0.33) to their year-6 and year-9 estimates. We then averaged these, together with their year-6 estimate, to construct a 4-7-year estimate. To align our 8-11-year estimates with their estimates for years 6, 9, and >=10, we first observed that the >=10 category corresponded to 11.5 years following graduation on average. This is because, among the cohorts in BGK's study observed for at least 10 years after graduation, the 1996 cohort was observed just at year 10; the 1995 cohort was observed just at years 10 and 11; and so on, with the 1990 cohort being observed at all years 10-16. Averaging the years of observation for each of the 1990-1996 graduating cohorts yielded an average year of observation, conditional on being observed at least 10 years since graduation, of 11.5. We constructed a pseudo-year-8 estimate by applying weights of (0.33,0.67) to BGK's year-6 and year-9 estimates. We constructed a pseudo-year-10 estimate by applying weights of (0.6,0.4) to their year-9 and year-11.5 estimates. We constructed a pseudo-year-11 estimate by applying weights of (0.2,0.8) to their year-9 and year-11.5 estimates. We then averaged these, together with their year-9 estimate, to construct an 8-11-year estimate.

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